New algorithms for accurate and efficient de-novo genome assembly from long DNA sequencing reads

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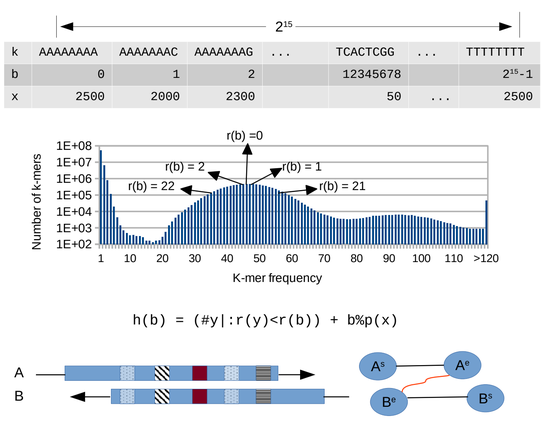
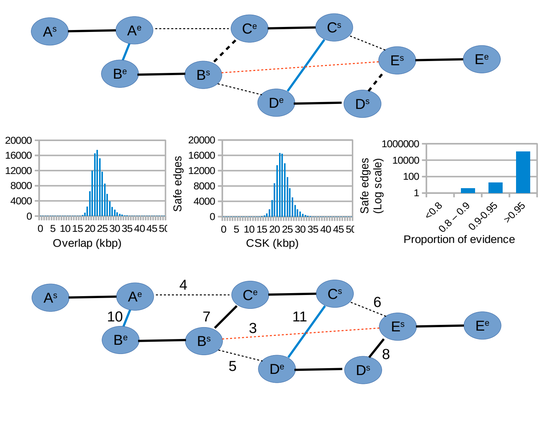
Introduction

Producing de-novo genome assemblies for complex genomes is possible thanks to long-read DNA sequencing technologies. However, maximizing the quality of assemblies based on long reads is a challenging task that requires the development of specialized data analysis techniques. Most of the commonly used tools to assemble long-read datasets implement the overlap-layout-consensus (OLC) algorithm. In this work we introduce a new software implementation for genome assembly from long-read sequencing data. It includes new algorithmic approaches to build overlap-layout-consensus (OLC) assembly graphs, and to identify layout paths. Benchmark experiments on PacBio HiFi and Nanopore data from organisms of different species including Escherichia coli, yeast, Drosophila melanogaster, rice, maize, and human show that our algorithms are competitive and, in some cases, more accurate, compared to previous solutions. These algorithms are implemented as part of the Next Generation Sequencing Experience Platform (NGSEP), allowing a tight integration with genome comparison and detection of genomic variants within a single easy-to-use tool for analysis of both short and long read DNA sequencing data.

Algorithm for OLC genome assembly

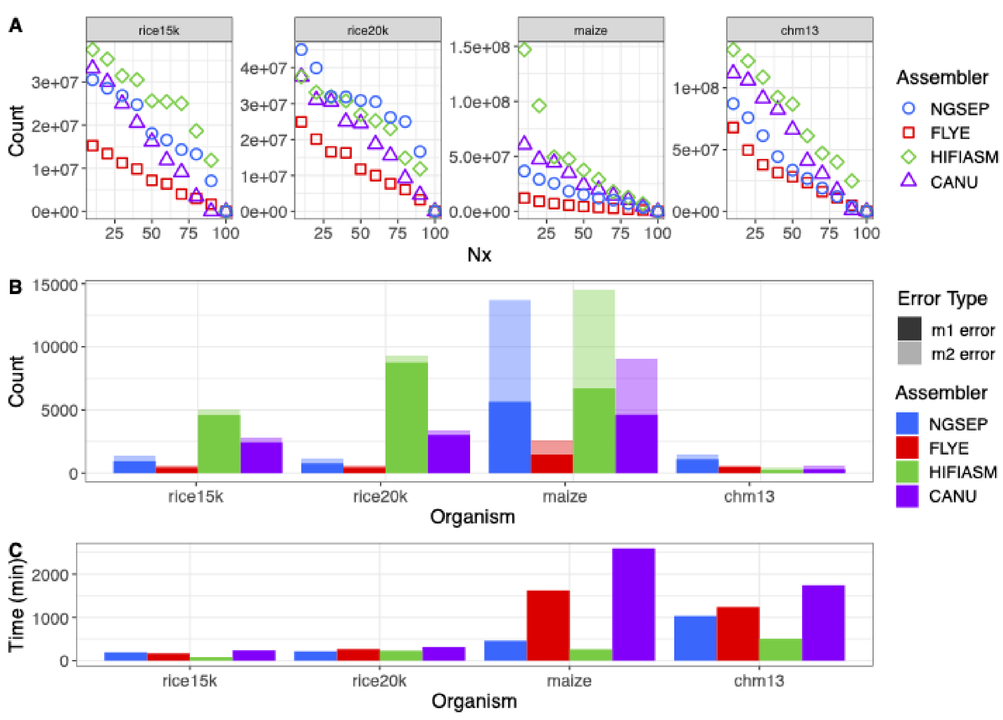
- The graph construction has two vertices for each read representing the start (5'-end) and the end (3'-end) of each read.
- We build a 15-mer spectrum table, calculating the count distribution across the reads using a fixed amount of 2Gb of RAM. This table defines a hashing scheme to select minimizers.
- The layout algorithm ranks and selects edges based on the knowledge that can be inferred from the distribution of the different statistics.
- We integrated the ReFHap algorithm for phased assembly of diploid samples





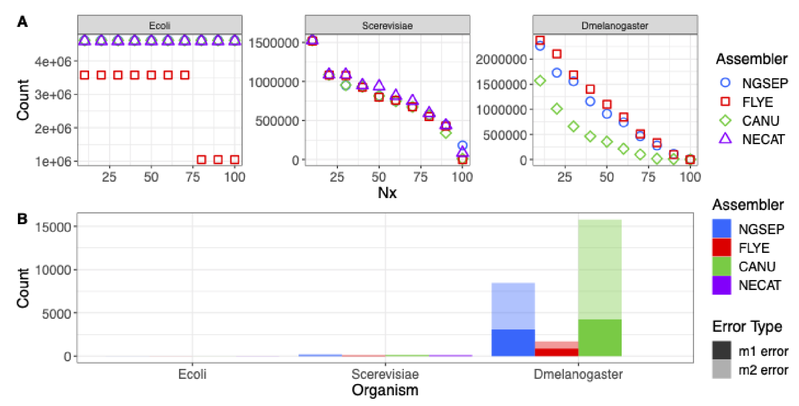
Benchmark with PacBio HiFi data

- We performed benchmark experiments against Flye, HiFiAsm and Canu on HiFi data from rice and maize accessions, and data from the human cell line CHM13
- Our assembler is competitive in both accuracy and computational efficiency



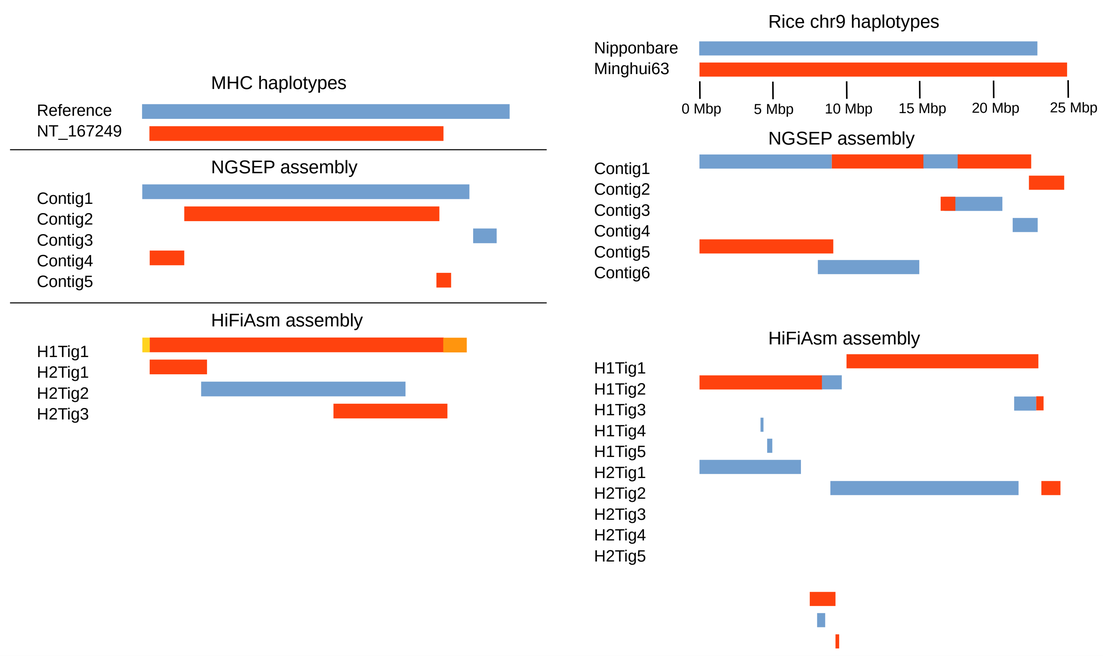
Benchmark with Oxford Nanopore data

- We performed benchmark experiments against Flye, and NECAT on Oxford nanopore data from *E. coli*, yeast and *Drosophila* samples
- After correction with NECAT, our assembler generates contiguous assemblies



Simulations of diploid individuals

- We simulated diploid individuals combining MHC alleles and combining rice chr9 assemblies



Phased assembly of HG002

- We assembled publicly available reads of the human individual HG002.
- Although our assembly has less contiguity, it recovers two alleles for a larger number of genes, compared to HiFiAsm

Metric	NGSEP	HiFiAsm phased	HiFiAsm primary
Length (Mbp)	5,593.63	5,979.17	3,109.3
NGA50 (Mbp)	1.68	91.07	69.24
Single copy genes in both the reference and the assembly	15,638	23,045	34,793
Single copy genes duplicated in the assembly	13,183	8,484	156
Single copy genes with exons mapped in different contigs	209	5	7
Single copy genes with 50%-99% of sequence mapped	674	93	67
Single copy genes with 10% - 50% of sequence mapped	477	23	3
Single copy genes with < 10% of sequence mapped	5,366	3,897	521
Duplicated genes in the reference found in the assembly	2,226	1,551	1,422
Total alleles of duplicated genes	7,499	8,187	8,266
Fraction of missing multicopy genes	0.61	0.73	0.62
Gene completeness (asmgene) (%)	81	89	98
Execution time (h)	49	52	52

Conclusions

- We contribute new alternatives to identify solutions to the genome assembly problem, facilitating de-novo construction of genome assemblies using long reads.
- Experiments with a wide variety of datasets indicate that our approach achieves competitive accuracy and efficiency, compared to state-of-the-art tools.
- New algorithms implemented in existing or novel tools could significantly improve the accuracy of phased assemblies directly from long reads.

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